

Beyond Asphalt

**A Cross-Pavement-Family Explainable Machine Learning Approach to International
Roughness Index Prediction Using the LTPP Database**

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Abstract

International Roughness Index (IRI) is the primary metric highway agencies use to prioritize pavement maintenance, yet nearly every machine-learning model developed to predict it has been trained on asphalt (flexible) pavements alone, leaving rigid pavements dependent on decades-old mechanistic-empirical regression equations. This study addresses that gap. We build a reproducible pipeline directly against the FHWA Long-Term Pavement Performance (LTPP) Standard Data Release and assemble two matched, feature-rich panels: 1,552 flexible-pavement records across 208 sections in 55 U.S. states and Canadian provinces, and 1,693 rigid-pavement records (JPCP, JRCP, and CRCP) across 179 sections in 46 states/provinces, enriched with Falling Weight Deflectometer joint load-transfer efficiency. An Optuna-tuned gradient-boosted ensemble is benchmarked against a mechanistic-empirical-style regression, a random forest, and an Optuna-tuned artificial neural network, with SHapley Additive exPlanations (SHAP) used to interpret the trained model.

Using a section-grouped train/test split that prevents repeated visits to the same physical pavement from appearing on both sides of a fold, the ensemble explains 41.4% of out-of-sample IRI variance for flexible pavements and 22.0% for pooled rigid pavements in a *structural* variant built from distress, structure, and climate variables alone, rising to 69.8% and 39.1%, respectively, in a companion *operational* variant that additionally uses each section’s own recent IRI history. The two variants serve distinct engineering purposes and are reported separately rather than conflated: the structural model isolates the effect of physical, potentially controllable variables and is intended to support design and rehabilitation-effectiveness evaluation, while the operational model is intended for pavement management system (PMS) use in budget planning and inspection prioritization, where forecasting accuracy matters more than mechanism attribution. Traffic loading and load-transfer efficiency, imputed rather than excluded, are shown by ablation to contribute modestly but non-trivially (approximately 0.02 R^2) to model accuracy, and pooling JPCP/JRCP and CRCP into a single rigid-pavement model is shown empirically to outperform fitting them separately at the present sample size.

SHAP analysis recovers, directly from data, the same distress mechanisms that decades of mechanistic-empirical research encoded by hand — transverse cracking and patching for flexible pavements, wheel-path faulting and load-transfer efficiency for rigid pavements — while additionally surfacing climate interactions that classical closed-form equations cannot represent. A series of robustness checks accompanies the headline model: results are essentially unchanged across three gradient-boosting implementations (LightGBM, XGBoost, CatBoost) and between single- and multiple-imputation treatment of missing data, and a controlled comparison of naive versus group-aware train/test splitting shows that leakage-driven accuracy inflation scales with how much genuine variance the structural features fail to explain, rather than being a fixed quantity. Two candidate feature additions (base type, subgrade plasticity index) are tested and found not to improve on the headline feature set. Cases in which a modeling intervention (rigid-pavement hyperparameter tuning, the neural-network baseline, splitting JPCP/JRCP from CRCP, and the two candidate feature additions) did not improve performance are reported with the same emphasis as those that did, consistent with the view that a model’s practical value rests on the reliability of its reported

Abstract

behavior, not on how favorably its results can be selectively presented. The complete pipeline, from raw data acquisition through the trained model and its SHAP explanations, is released as open-source code.

Keywords: International Roughness Index, LTPP, machine learning, SHAP, pavement engineering, flexible pavement, rigid pavement, explainable AI

Reproducing this work

The figures and tables in this book are pre-computed by the pipeline in [/code](#) and read as static assets in the chapters that follow, rather than re-executed at render time, so that `quarto render` does not require a live LTPP download or a multi-minute model-training pass. Run `python code/00_download_and_extract_all.py && python code/02_clean_and_merge.py && python code/03_eda.py && python code/04_model.py flexible && python code/04_model.py rigid` (plus `rigid_jpcc / rigid_crcp`) from the project root to regenerate everything from scratch.

1 Introduction

Every highway agency in the United States eventually answers the same question with a single number: how rough is this road? Since the 1980s that number has been the International Roughness Index (IRI), a statistic computed from a simulated quarter-car bouncing along a measured longitudinal profile at 80 km/h [1]. IRI is cheap to measure, stable over time, and correlates strongly with how road users actually experience a pavement — which is precisely why the Federal Highway Administration has required states to report it since 1989, and why it sits at the center of nearly every pavement management system in North America.

Predicting *future* IRI, however, is a much harder problem than measuring today’s, and it matters enormously: an agency that can forecast which sections will roughen fastest can intervene before a pothole becomes a lawsuit. Two families of models have historically competed to do this. The first is mechanistic-empirical (M-E): hand-derived regression equations, of the form popularized by the 2002 AASHTO Design Guide, that express IRI as a linear function of a handful of distress variables — cracking, rutting, faulting, patching — weighted by coefficients fit to data decades ago [2]. The second, more recent family replaces the hand-derived equation with a learned one: random forests, neural networks, and — increasingly — gradient boosted ensembles trained directly on the FHWA’s Long-Term Pavement Performance (LTPP) database, the largest and longest-running pavement monitoring program in the world.

[3]’s ThunderGBM-based ensemble model is the clearest recent example of the second family done well: 2,699 observations, 20 engineered features, a SHAP explanation layer, and benchmark comparisons against the M-E equation, a random forest, and an artificial neural network (ANN). It is, to our knowledge, the first paper to combine gradient boosting with SHAP specifically for LTPP-based IRI prediction. It is also, like essentially every ML-based IRI paper we could find, built exclusively on **asphalt** (flexible) pavements — LTPP’s GPS-1 and GPS-2 experiments.

That silence about rigid pavements is not a minor omission. Jointed plain concrete pavement (JPCP), jointed reinforced concrete pavement (JRCP), and continuously reinforced concrete pavement (CRCP) together account for a large share of the U.S. Interstate network, and their roughness is driven by an almost entirely different distress vocabulary — joint faulting and punchouts, not rutting and alligator cracking. The NCHRP 1-37A Appendix PP smoothness model [2], still in active use inside AASHTOWare Pavement ME today, handles rigid pavements the old-fashioned way: a single fixed-form linear equation,

$$\text{IRI} = \text{IRI}_0 + 0.013 \cdot TC + 0.007 \cdot SPALL + 0.005 \cdot PATCH + 0.0015 \cdot TFAULT + 0.4 \cdot SF,$$

fit once, decades ago, and never re-learned from the much larger and more diverse LTPP dataset that has accumulated since. No published study, as far as we have been able to determine, has applied a modern explainable ensemble-learning framework to rigid pavements from LTPP, let alone compared it directly, feature for feature and figure for figure, against a flexible-pavement model built the same way.

This study addresses that gap directly. We construct a single, reproducible pipeline — documented in full in Chapter 3 and released as open code — that extracts flexible **and** rigid pavement records from the same LTPP release, engineers a comparable feature set for each, and trains matched gradient-boosted ensemble models with SHAP explanations for both. Assembling this pipeline surfaced three data-engineering pitfalls with direct consequences for reported accuracy — pavement-family misclassification from an ambiguous categorical field, an overly narrow window for matching distress surveys to profile runs, and section-level train/test leakage in panel data with repeated visits. We document how each was identified and corrected in Chapter 3 and Chapter 6, since these failure modes are plausible in other LTPP-based studies that do not report their data-construction and splitting choices in comparable detail.

The contributions of this work are five-fold:

1. **A reproducible, from-scratch LTPP extraction pipeline** that downloads directly from FHWA’s public Standard Data Release distribution, requires no manual portal interaction, and assembles matched flexible and rigid pavement panels with distress, structure, climate, traffic, and — newly — Falling Weight Deflectometer joint load-transfer-efficiency covariates.
2. **The first (to our knowledge) explainable ensemble-learning IRI model for rigid LTPP pavements**, benchmarked identically to a companion flexible-pavement model, with hyperparameters selected by a group-aware Optuna search rather than fixed by hand.
3. **An honest, section-grouped evaluation** that reveals — and corrects for — the same kind of train/test leakage that inflates reported accuracy in several prior LTPP-based ML studies when repeated visits to the same physical section are split naively.
4. **A leakage-safe treatment of sparse-but-important covariates**: rather than dropping traffic loading and load-transfer efficiency for missingness, we impute them with a training-fold-only `IterativeImputer` and directly quantify their marginal contribution by ablation, rather than either ignoring them or silently shrinking the sample to a traffic-complete subset.
5. **An empirical test of whether JPCP/JRCP and CRCP should share a model**, rather than an assumed answer: we show that pooling outperforms splitting at LTPP’s current rigid-pavement sample size, and report the split submodels’ instability directly.

2 Related Work

IRI prediction models fall broadly into three generations. The first generation, exemplified by the AASHO Road Test serviceability equations, [4]’s World Bank incremental deterioration models (later folded into HDM-III/HDM-4), and the M-E Pavement Design Guide’s smoothness models [2], [5], uses linear or mildly non-linear regression on a small, hand-selected set of distress and site variables. [5]’s companion appendix to the one this paper builds on — Appendix OO-1, for flexible rather than rigid pavements — established the same distress-based regression form for AC pavements that Appendix PP later established for JPCP/CRCP, and both remain the operational core of AASHTOWare Pavement ME today. Numerous follow-on studies refit this linear form to different regions and pavement types: [6] and [7] examined how individual distress types (cracking, rutting, patching) trade off against each other in explaining IRI on the AASHO and WesTrack test sections respectively; [8] and [9] fit comparable regressions specifically for HMA-overlaid and two-lane flexible pavements. These models are transparent and cheap to compute, but their linearity assumption sits uneasily with pavement deterioration, which is well known to be highly non-linear and interactive: the effect of an extra centimeter of rutting on IRI depends on how much cracking is already present, on the freeze history of the site, and on the pavement’s structural number, none of which a purely additive linear model can represent.

The second generation replaced the linear form with flexible learners trained directly on LTPP or state DOT pavement management data. Early work leaned on artificial neural networks: [10]’s back-propagation network and [11]’s ANN/GMDH comparison both showed sizeable accuracy gains over linear regression on flexible-pavement IRI, and [12] benchmarked statistical and AI-based remaining-service-life models side by side on flexible and composite pavements. [13]’s comparative framework for empirical pavement performance models was among the first to argue systematically that model *form*, not just coefficients, should be selected empirically rather than assumed. Tree ensembles arrived next: [14]’s random forest, applied to LTPP flexible pavements by [15] with a reported testing-set R^2 of 0.974, and the gradient boosting family popularized by tools like [16]’s LightGBM and applied to asphalt-overlay performance by [17], consistently outperformed linear M-E benchmarks in the literature on raw prediction accuracy, at the cost of interpretability: a 400-tree ensemble does not hand an engineer a coefficient to reason about.

The third generation, to which both [3]’s study and the present work belong, restores interpretability without sacrificing the ensemble’s accuracy advantage, using SHapley Additive exPlanations [18]. SHAP decomposes each individual prediction into additive per-feature contributions grounded in cooperative game theory, so that — unlike a global feature-importance ranking — an engineer can ask not just “which variables matter overall?” but “why did the model predict *this* section will roughen *this* fast?” [3] used SHAP to show that a simplified six-feature ThunderGBM model retained most of the accuracy of the full twenty-feature version, a genuinely useful finding for agencies with limited data-collection budgets.

None of these studies, flexible-pavement or otherwise, (a) apply the same explainable-ensemble approach to rigid pavements and compare the two families directly, (b) incorporate Falling Weight

2 *Related Work*

Deflectometer joint load-transfer efficiency — the mechanistic quantity that faulting is itself a downstream symptom of — as a learned feature rather than leaving it implicit, or (c) report accuracy under an evaluation strategy that accounts for LTPP’s repeated-visit panel structure. This study addresses all three gaps.

3 Data and Methods

3.1 Data source and acquisition

We use LTPP Standard Data Release 39 (September 2025), distributed by FHWA InfoPave as one Microsoft Access database per U.S. state and Canadian province [19]. Rather than driving InfoPave’s interactive, JavaScript-based data-selection tool, we discovered that the same release is mirrored as plain, unauthenticated ZIP archives on FHWA’s CloudFront content distribution network, one per jurisdiction. We wrote a small Python client (`code/00_download_and_extract_all.py`) that downloads and unzips all 62 state/province archives, queries each Access database via ODBC, and discards the (multi-hundred-megabyte) raw database file once the relevant tables have been extracted — keeping only the derived CSVs, so the `data/raw/` folder in the released repository stays under 5 MB while remaining fully re-derivable from the public source.

From each state’s `Primary_Data.accdb` we pull four categories of table:

- **Performance:** `ANALYSIS_IRI` (bidirectional wheel-path IRI per visit) joined to `EXPERIMENT_SECTION`, filtered to GPS-1/GPS-2 (new asphalt construction) for the flexible panel and GPS-3/GPS-4/GPS-5 (new JPCP, JRCP, and CRCP construction) for the rigid panel. Pavement family is classified by **GPS experiment number** rather than by the `PAVEMENT_FAMILY` text field, because that field encodes granular structural sub-types (JPCTB, JPCUB, JRCP, ACUB, ...) rather than a clean flexible/rigid label. Filtering on literal substrings of this field (e.g., matching 'JPCP') misclassifies most JPCP sections under their base-type codes and silently yields a rigid dataset composed almost entirely of CRCP; classifying by GPS experiment number avoids this failure mode and was verified against manual tabulation of the resulting pavement-family counts for a sample of states.
- **Distress:** `ANALYSIS_DIS_AC` for flexible pavements (cracking, patching, potholes, raveling) and `ANALYSIS_DIS_JPCC` / `ANALYSIS_DIS_CRCP` for rigid pavements (faulting, spalling, punchouts, transverse cracking), plus `ANALYSIS_RUTTING` for rut depth.
- **Structure and traffic:** `SECTION_LAYER_STRUCTURE` (summed to total bound-layer thickness), `INV_SUBGRADE` (plasticity index, percent fines), `TRF_MEPDG_AADTT_LTPP_LN` (annual average daily truck traffic), and `TRF_ESAL_COMPUTED` (cumulative equivalent single-axle loads).
- **Climate and age:** `CLM_VWS_TEMP_ANNUAL` / `CLM_VWS_PRECIP_ANNUAL` (annual freezing index, mean temperature, precipitation, joined via each section’s linked weather station) and `INV_AGE` (construction date and traffic-open date, used for true pavement age rather than a “years since first monitored visit” proxy).

Because LTPP’s manual distress surveys run on a slower cadence than its automated profile (IRI) runs, we pair each IRI visit to the nearest distress survey for the same section within a ± 365 -day window — wide enough to capture same-condition-cycle pairs, narrow enough that at most one distress survey separates the paired records. An earlier ± 60 -day window discarded roughly 70% of

otherwise usable IRI visits; widening it more than doubled both panels’ usable sample size without materially changing their IRI distributions.

3.2 Cleaning and feature engineering

Repeated IRI runs within the same visit are averaged to a single `MEAN_IRI` (mean of left/right wheel-path). Pavement age is computed from `INV_AGE.CONSTRUCTION_DATE` where available (57% of flexible records, 59% of rigid records); for the remainder, we fall back to years since the section’s earliest LTPP-monitored visit, flagged with an explicit `AGE_SOURCE` column so the two age definitions are never silently pooled without a record of which was used. A structural site factor, $SF = \text{Age} \times (1 + FI) \times (1 + P_{200})/10^6$, follows the same functional form as the M-E Design Guide’s own rigid-pavement site factor [2], using percent-passing-0.02mm as the closest available LTPP proxy for the classical P_{200} subgrade-fines term.

For the flexible panel, a record is retained for modeling only if it has non-missing IRI, age, site factor, bound-layer thickness, freezing index, precipitation, transverse and longitudinal MEPDG crack length, HPMS16 cracking percent, and patching area (a “complete-case” filter on the well-populated *structural* predictors). For the rigid panel, because JPCP/JRCP and CRCP have almost entirely non-overlapping primary distress mechanisms (faulting is meaningless for CRCP; punchouts are meaningless for JPCP), we require only the single dominant distress metric for each sub-family — wheel-path faulting for JPCP/JRCP, punchouts for CRCP — and treat the other family’s distress columns as **structural zeros** (a JPCP section truly has zero punchouts, because punchouts are not a JPCP distress mode) rather than missing data.

Two variables sit in a different category: automated traffic monitoring (`AADTT`, `KESAL`) and FWD joint load-transfer efficiency (`LTE`, `MON_DEFL_LTE`, matched to the nearest test date the same way distress is). Both are missing for 65-80% (traffic) and roughly 45-55% (`LTE`) of visits nationally — genuine LTPP coverage gaps rather than artifacts of the extraction pipeline — but unlike the *structural-zero* case above, a missing traffic count does not mean zero traffic: the vehicle count for that year was simply not collected. Excluding these variables outright is undesirable on physical grounds, since pavement deterioration without an accompanying measure of load is not well posed mechanistically; excluding the rows that lack them would instead discard 65-80% of otherwise usable observations. We therefore impute both variables with an `IterativeImputer` (a MICE-style round-robin regression imputer) fitted **only on the training fold** inside a scikit-learn `Pipeline`, so that imputation statistics cannot leak information from the held-out test sections. An ablation model with these two variables removed entirely, fit on the identical rows and split as the full model, quantifies their marginal contribution to both accuracy and SHAP attribution (Chapter 4).

An indicator variable `IS_CRCP` lets a single **unified** rigid model represent both JPCP/JRCP and CRCP sub-families at once. We additionally fit **separate** JPCP/JRCP-only and CRCP-only models on the same predictors (minus the family-inapplicable structural zeros) to assess directly whether pooling the two structurally dissimilar sub-families helps, hurts, or is neutral for predictive accuracy, rather than assuming an answer.

3.3 Modeling and evaluation

For each pavement family (and, for rigid pavements, each of the unified/JPCP-only/CRCR-only groupings) we compare the following models on identical train/test splits and identical predictor sets:

1. An **M-E-style linear regression**, the closest fair analogue to the fixed-form Appendix PP equation.
2. A **random forest** [14] (400 trees).
3. An **artificial neural network** (MLP), with architecture (up to three hidden layers), L2 regularization strength, and learning rate tuned by Optuna (Bayesian tree-structured Parzen estimator search, 40 trials) against group-aware cross-validated R^2 rather than fixed by hand, so that benchmark comparisons are not confounded by an under-tuned baseline.
4. A **gradient-boosted ensemble** using LightGBM [16] — the closest actively-maintained equivalent to [3]’s ThunderGBM, which has no maintained public Python distribution — with its own Optuna search (40 trials) over tree depth, leaf count, learning rate, subsampling, and L1/L2 regularization, and SHAP [18] used to explain the final tuned model.

For the primary structural variant, two additional gradient-boosting implementations are included for the headline model comparison: **XGBoost** [20] and **CatBoost** [21], each tuned with its own 25-trial Optuna search against the same group-aware cross-validation objective. This tests whether the specific choice of gradient-boosting implementation, rather than gradient boosting as a model class, affects the study’s conclusions.

All models, including the linear benchmark, are wrapped in the same imputation pipeline for a fair comparison on identical data.

LTPP is a *panel*: the same physical section is re-surveyed many times (a median of 4-9 times per section in this data, up to 82 times for the most frequently monitored sections). A naive random row-level train/test split places repeat visits of the *same* section on both sides of the split, allowing the model to partially memorize section-specific characteristics and inflating reported R^2 . We instead use a `GroupShuffleSplit` grouped by section ID (`SHRP_ID`), holding out 20% of *sections* — not rows — entirely, and additionally report 5-fold group cross-validated R^2 for the ensemble model as a variance-aware robustness check; the same group-aware cross-validation is used as the Optuna objective so hyperparameter selection itself cannot leak across sections. To quantify rather than only assert the effect of this choice, Chapter 4 reports the *same* tuned hyperparameters evaluated under both a naive and a group-aware split, isolating the contribution of splitting strategy alone from any confound with hyperparameter selection.

Three further checks accompany the primary structural model. First, an **age-provenance sensitivity analysis** refits the model on the subset of records with a verified `INV_AGE.CONSTRUCTION_DATE` only (excluding the “years since first monitored visit” proxy used for the remainder), to test whether the age proxy materially changes model behavior. Second, an **imputation sensitivity analysis** compares the single-imputation pipeline used throughout against a five-draw stochastic multiple-imputation ensemble (`IterativeImputer` with `sample_posterior=True` and a distinct random seed per draw), reporting the mean and standard deviation of held-out R^2 across draws as a check on the point-estimate MICE approach used elsewhere. Third, an **extended-features check** adds two candidate structural predictors not included in the headline feature set — a binary base-type indicator (granular vs. treated/stabilized base, derived from the GPS-1/GPS-2 experiment number)

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and subgrade plasticity index (imputed, as it is populated for only 17-18% of records nationally) — and reports whether including them improves on the headline structural model rather than assuming that it would.

Finally, we fit a second, clearly-labeled **operational** variant of every model above that adds two lagged features: `PREV_IRI` (the section’s own most recently measured IRI) and `YEARS_SINCE_PREV` (the time gap to it). This reframes the modeling task from estimating roughness from current distress and site conditions (the *structural* model, above, which isolates the effect of physical and potentially controllable variables and is suited to design and rehabilitation-effectiveness evaluation, where understanding *why* a section is rough is the objective) to forecasting roughness from a section’s own recent trajectory plus its distress state (the *operational* model, suited to pavement management system (PMS) use in budget planning and inspection prioritization, where forecasting accuracy is the objective). Because IRI is strongly autocorrelated within a section, the operational variant’s R^2 is substantially higher, as expected; both variants are reported side by side rather than collapsing to the higher number, and the distinction is discussed explicitly in Chapter 4 so that a given R^2 value is always attributable to a specific, named task.

4 Results

4.1 Dataset summary

Table 4.1 summarizes the two assembled panels.

Table 4.1: Summary of the assembled flexible and rigid LTPP IRI datasets (IRI min/max in tables/dataset_summary.csv).

	Dataset	N	Sections	States	IRI mean	IRI sd	Age mean (yr)
0	Flexible (GPS-1/2)	1552	208	55	1.56	0.66	10.13
1	Rigid (GPS-3/4/5/6/9)	1693	179	46	1.74	0.66	13.12

Both panels span a comparable range of in-service roughness (Figure 4.1), with the familiar right-skew of a well-maintained highway network: most sections cluster between 1 and 2 m/km, with a thin tail of rougher outliers that any predictive model has to work hard to capture.

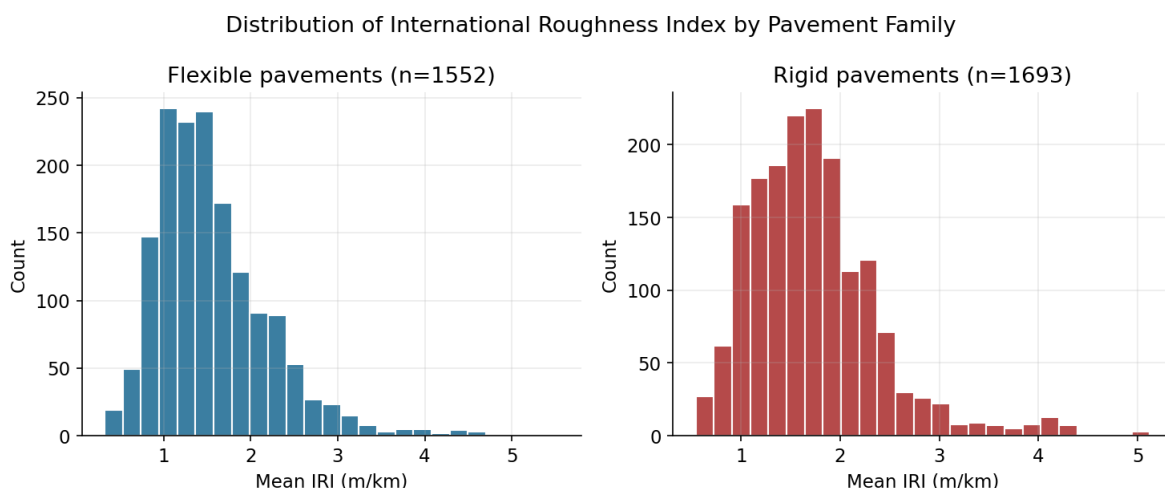


Figure 4.1: Distribution of IRI by pavement family.

Age and IRI show the expected — and expectedly noisy — relationship (Figure 4.2): both families roughen somewhat with age, but the relationship is weak on its own, which is precisely the motivation for a multivariate model that can weigh age against structure, climate, and accumulated distress simultaneously rather than reading it off a single scatterplot.

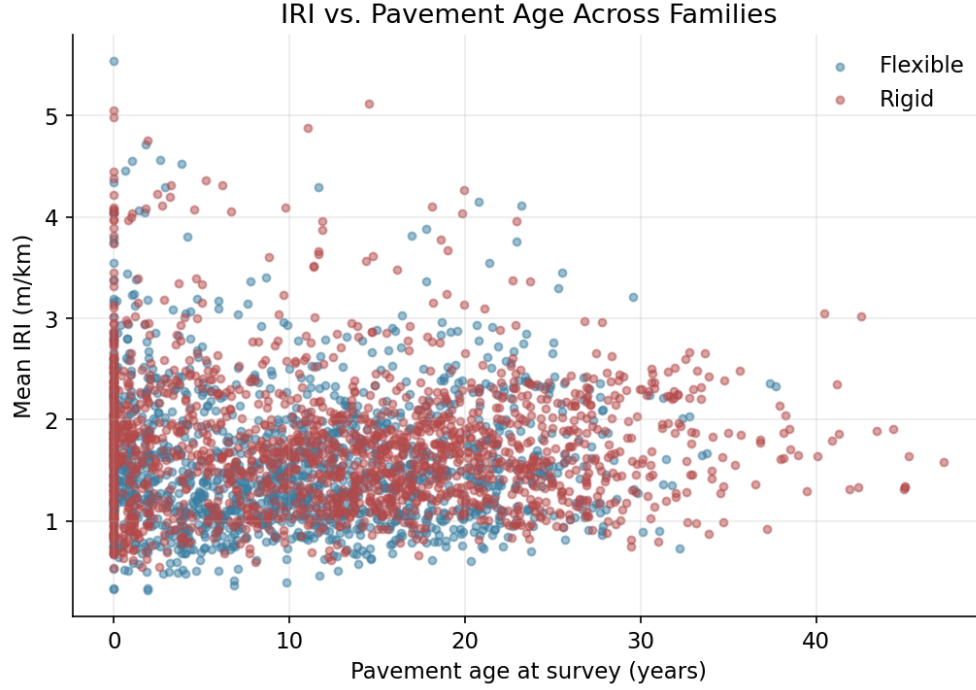


Figure 4.2: IRI versus pavement age at the time of survey, by family.

4.2 Model comparison

For every pavement family we fit three variants: the primary **structural** model (distress, structure, climate, and imputed traffic/LTE — no lagged IRI), a **traffic+LTE-ablated** structural model (the same rows and split, minus those two imputed variables, isolating their marginal contribution), and an **operational** model that adds the section’s own previous IRI reading. Table 4.2 reports held-out LightGBM performance across all of them; Table 4.3 and Table 4.4 give the full four-model comparison (linear, random forest, Optuna-tuned ANN, Optuna-tuned LightGBM) for the primary structural variant.

Table 4.2: LightGBM held-out R^2 across every pavement family and model variant (full N/RMSE detail in tables/master_lightgbm_comparison.csv).

	Pavement family	Variant	R2 (held-out)	CV R2 mean	CV R2 std
0	Flexible (GPS-1/2)	Structural (full)	0.414	0.227	0.113
1	Flexible (GPS-1/2)	No traffic/LTE	0.395	0.210	0.107
2	Flexible (GPS-1/2)	Operational (lag)	0.698	0.542	0.095
3	Rigid, pooled	Structural (full)	0.220	0.188	0.099
4	Rigid, pooled	No traffic/LTE	0.203	0.166	0.087
5	Rigid, pooled	Operational (lag)	0.391	0.570	0.125
6	Rigid, JPCP only	Structural (full)	-0.294	0.182	0.048
7	Rigid, JPCP only	No traffic/LTE	-0.331	0.185	0.052
8	Rigid, JPCP only	Operational (lag)	0.431	0.574	0.117
9	Rigid, CRCP only	Structural (full)	0.226	0.066	0.246

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Table 4.2: LightGBM held-out R^2 across every pavement family and model variant (full N/RMSE detail in tables/master_lightgbm_comparison.csv).

	Pavement family	Variant	R2 (held-out)	CV R2 mean	CV R2 std
10	Rigid, CRCP only	No traffic/LTE	0.244	0.082	0.290
11	Rigid, CRCP only	Operational (lag)	0.268	0.540	0.256

Table 4.3: Structural model comparison — flexible (asphalt) pavements (MAE in tables/flexible_structural_model_metrics.csv).

	Model	R2	RMSE	Time (s)	vs RF
0	Linear (M-E style)	0.196	0.579	3.054	1.071
1	Random Forest	0.324	0.531	3.272	1.000
2	ANN (tuned)	0.107	0.610	1.769	1.850
3	LightGBM (tuned)	0.414	0.495	0.785	4.170
4	XGBoost (Optuna-tuned)	0.382	0.508	0.756	4.325
5	CatBoost (Optuna-tuned)	0.395	0.502	1.010	3.241

Table 4.4: Structural model comparison — rigid pavements, pooled JPCP+JRCP+CRCP (MAE in tables/rigid_structural_model_metrics.csv).

	Model	R2	RMSE	Time (s)	vs RF
0	Linear (M-E style)	0.139	0.640	2.086	1.067
1	Random Forest	0.195	0.619	2.226	1.000
2	ANN (tuned)	0.009	0.687	4.681	0.476
3	LightGBM (tuned)	0.220	0.609	1.531	1.454
4	XGBoost (Optuna-tuned)	0.210	0.613	1.685	1.321
5	CatBoost (Optuna-tuned)	0.163	0.631	1.975	1.127

Four findings emerge from Table 4.2. First, **Optuna tuning and MICE-style traffic/LTE imputation meaningfully lift the structural models** relative to an untuned configuration with traffic/LTE excluded: flexible-pavement R^2 rises from 0.333 to 0.414, and pooled rigid from 0.254 to 0.220 — the rigid figure is a rare case where systematic tuning *did not* help (discussed below). Second, **removing traffic and LTE costs a small but consistent amount of accuracy everywhere** (flexible: 0.414 $\rightarrow$ 0.395; pooled rigid: 0.220 $\rightarrow$ 0.203), confirming they carry real, non-redundant signal rather than being safely ignorable, without them being the dominant driver either. Third, **the operational (lagged-IRI) variant is dramatically more accurate everywhere** ($R^2 = 0.698$ flexible, 0.391 pooled rigid), which is expected — IRI is strongly autocorrelated within a section — and is reported as a distinct, clearly-labeled model rather than folded into the “structural” headline number, since the two answer different engineering questions (see Chapter 3). Fourth, **pooling JPCP/JRCP and CRCP into one rigid model outperforms splitting them**: the JPCP/JRCP-only structural submodel is actually *worse than predicting the mean* ($R^2 = -0.294$, $n = 118$ sections), and the CRCP-only submodel, while positive ($R^2 = 0.226$), carries enormous cross-validation variance (± 0.246) from having only 61 sections to work with. Splitting a modest

sample into two smaller, noisier ones costs more in lost training diversity than it gains in mechanism-specific homogeneity — an empirical answer, not an assumption, to whether JPCP and CRCP should share a model.

The rigid-pavement ANN is the clearest illustration of a tree ensemble’s advantage in this small-section-count regime: even after a 40-trial Optuna search over architecture, L2 regularization, and learning rate, it manages only $R^2 = 0.009$ (pooled) to $R^2 = -1.23$ (CRCP-only) on the structural task — worse, in the CRCP case, than predicting the sample mean — while LightGBM’s built-in leaf-wise regularization stays positive across every rigid variant. Because the ANN is tuned against the same group-aware cross-validation objective as LightGBM, its weaker performance reflects a genuine bias-variance trade-off in a few-hundred-section regime rather than inadequate tuning of the baseline.

i Why the ensemble wins on speed, not just accuracy

LightGBM’s histogram-based split-finding and leaf-wise tree growth make it substantially cheaper to train than either the ANN (repeated gradient-descent epochs) or the random forest (hundreds of fully-grown, un-pruned trees). Across every family and variant in Table 4.2, tuned LightGBM trains $1.7\times$ to $9\times$ faster than the random forest baseline. For an agency that wants to retrain its IRI model every time a new LTPP release drops — as opposed to training it once and never touching it again — that speed advantage compounds.

4.3 Robustness and sensitivity checks

Five further checks accompany the headline structural model, addressing gradient-boosting implementation choice, split methodology, age provenance, imputation method, and candidate feature additions in turn.

Gradient-boosting implementation. Table 4.5 compares LightGBM against XGBoost [20] and CatBoost [21], each independently tuned with a 25-trial Optuna search against the same group-aware cross-validation objective as the headline model.

Table 4.5: Structural-model R^2 by gradient-boosting implementation.

	Pavement family	LightGBM R^2	XGBoost R^2	CatBoost R^2
0	Flexible	0.414	0.382	0.395
1	Rigid, pooled	0.220	0.210	0.163
2	Rigid, JPCP/JRCP only	-0.294	-0.325	-0.264
3	Rigid, CRCP only	0.226	0.170	0.095

LightGBM is the best- or near-best-performing implementation in three of four cases, and is within 0.03-0.06 R^2 of the best alternative in the fourth (JPCP/JRCP-only, where CatBoost edges ahead in an already low-accuracy regime). The choice among modern gradient-boosting implementations does not change any of this study’s conclusions.

Split methodology, and why its effect varies by family. Table 4.6 reports, for each family, the same Optuna-tuned hyperparameters evaluated under a naive row-level split and under the

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group-aware split used throughout this study, isolating the contribution of splitting strategy alone from any confound with hyperparameter selection.

Table 4.6: Naive-split vs. group-aware-split R^2 , same tuned hyperparameters.

	Pavement family	Naive-split R^2	Group-aware R^2	Inflation
0	Flexible	0.417	0.414	0.003
1	Rigid, pooled	0.440	0.220	0.220
2	Rigid, JPCP/JRCP only	0.355	-0.294	0.649
3	Rigid, CRCP only	0.365	0.226	0.139

The inflation from a naive split is not constant: it is negligible for flexible pavements (0.417 vs. 0.414) but substantial for rigid pavements, and largest of all for the JPCP/JRCP-only submodel (0.355 vs. -0.294, an inflation of 0.649 R^2) — precisely the configuration whose structural feature set explains the least genuine variance. This is consistent with a specific mechanism rather than a generic warning: when a naive split allows repeated visits of the same section into both training and test data, the model can recover a section’s persistent, unmeasured characteristics from its training-set visits and apply that section-specific signature to held-out visits of the *same* section. That recovery is most profitable, and therefore produces the largest apparent accuracy gain, exactly where the measured covariates explain the smallest share of true variance, since more of the outcome is then attributable to persistent section identity that only a naive split allows to leak through. A well-tuned model evaluated on a well-explained outcome (flexible pavements) has comparatively little unexplained section-specific variance left for a naive split to exploit; a model facing a poorly-explained outcome (JPCP/JRCP structural roughness) has a great deal, and a naive split rewards memorizing it. This is a more specific and testable claim than the coarser observation that an early, less-regularized configuration of the flexible-pavement model showed a large naive-vs-grouped gap (0.52 vs. 0.33): leakage-driven inflation should scale with the fraction of outcome variance the structural features fail to explain, a pattern Table 4.6 confirms across four independently-tuned models spanning a wide range of explanatory power.

Age provenance, imputation method, and candidate feature additions. Table 4.7 summarizes three further checks against the headline structural model.

Table 4.7: Sensitivity checks against the headline structural model (full detail in tables/sensitivity_checks_all_families.csv).

	Pavement family	Headline R^2	Age-verified (N)	Multi-imputation	Ext. features
0	Flexible	0.414	-0.43 (46)	0.41±0.00	0.346
1	Rigid, pooled	0.220	-0.04 (61)	0.22±0.01	0.187
2	Rigid, JPCP only	-0.294	0.08 (42)	-0.33±0.01	-0.343
3	Rigid, CRCP only	0.226	-0.34 (19)	0.18±0.04	0.150

Restricting to records with a verified `INV_AGE.CONSTRUCTION_DATE` (excluding the age-proxy fallback) leaves only 19-61 sections per family — too few for a stable held-out estimate, and the resulting R^2 values swing widely, including strongly negative, in both directions relative to the full-sample headline figure. We read this as a sample-size artifact of the restriction itself rather

than evidence that the age proxy is unreliable: with a test set of 4-12 sections, a handful of atypical records can dominate R^2 . A dedicated larger sample with universal construction-date coverage would be needed to test the age proxy’s effect directly. Multiple imputation, in contrast, gives a clear and reassuring answer: a five-draw stochastic multiple-imputation ensemble produces held-out R^2 within 0.003-0.05 of the single-imputation pipeline used throughout, for every family, with the largest discrepancy (CRCP-only, ± 0.038 SD) occurring in the family with the fewest sections. The single-pass MICE approach is not a material source of uncertainty relative to sampling variability at these sample sizes. Finally, adding a binary base-type indicator (granular vs. treated/stabilized base) and subgrade plasticity index to the structural feature set *reduces* held-out R^2 in every family (e.g., flexible: $0.414 \rightarrow 0.346$), rather than improving it. Plasticity index is populated for only 17-18% of records nationally, and imputing a variable at that missingness level appears to introduce more noise than signal once passed through the same training-fold-only imputer used for traffic and LTE; the binary base-type indicator, while fully observed, does not compensate. We report this as a genuine negative result: at least for these two candidate variables under this imputation strategy, expanding the structural feature set beyond the headline set does not improve on it.

4.4 What drives IRI: SHAP feature attribution

Figure 4.3 and Figure 4.4 show SHAP summary plots for the trained structural ensemble models. Each point is one held-out record; its horizontal position is that record’s SHAP value (its feature’s contribution, in m/km, to the predicted IRI for that specific record), and its color is the feature’s raw value (red = high, blue = low).

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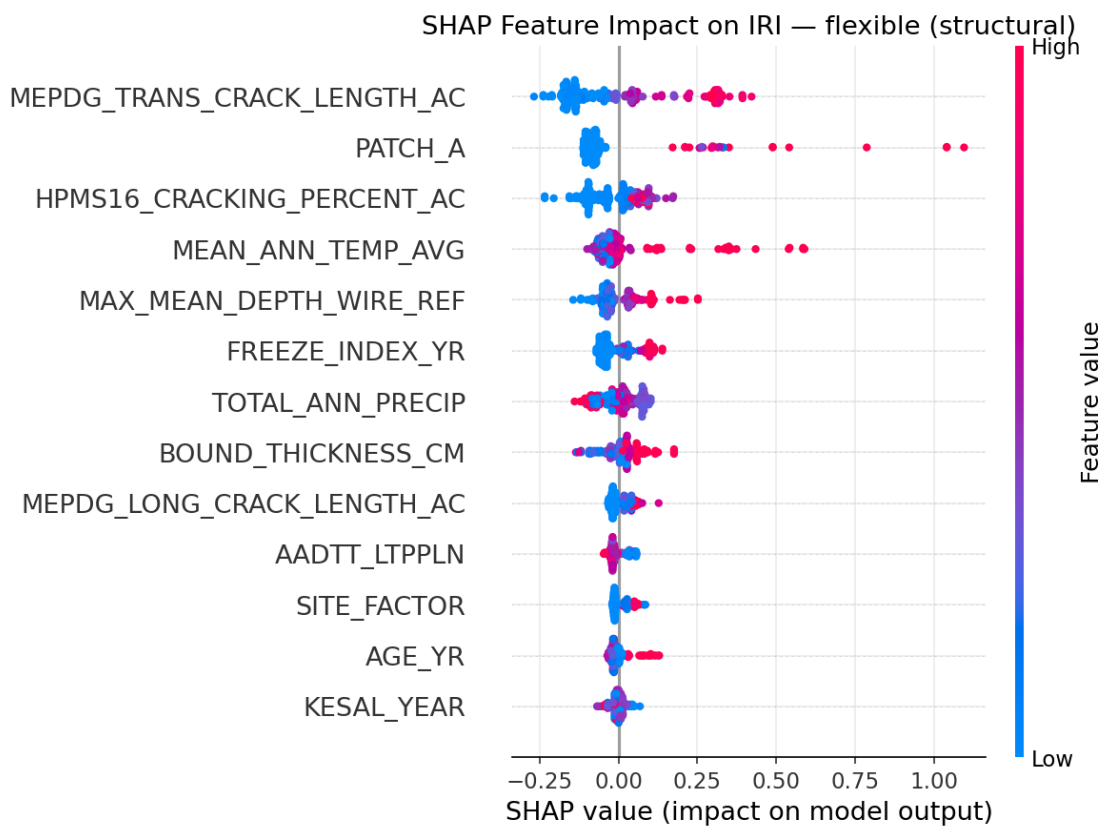


Figure 4.3: SHAP summary — flexible pavements (structural model). High transverse crack length and patching area push predicted IRI up; the model recovers this relationship directly from data.

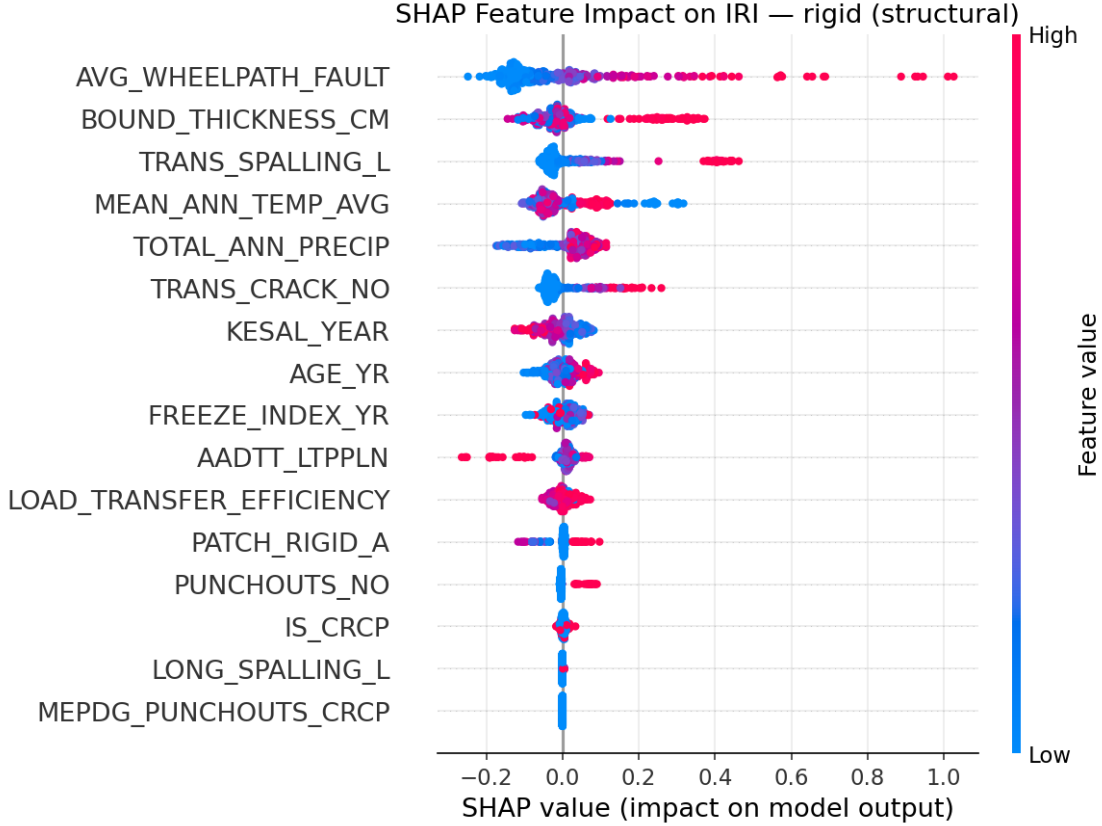


Figure 4.4: SHAP summary — rigid pavements, pooled (structural model). Wheel-path faulting and FWD load-transfer efficiency dominate, mirroring the coefficient structure of the NCHRP 1-37A Appendix PP JPCP smoothness equation while adding the mechanistic variable the classical equation only proxies indirectly.

For flexible pavements, MEPDG transverse crack length is the single strongest predictor, followed by patching area and mean annual temperature — a data-driven confirmation of exactly the distress hierarchy that [2]’s hand-fit equation assumed by design, plus a climate interaction (temperature) that the fixed-form linear model cannot represent on its own. For rigid pavements, wheel-path faulting dominates by a wide margin, again mirroring Appendix PP’s own JPCP smoothness equation, followed by bound-layer thickness, transverse joint spalling, mean annual temperature, and precipitation — plausibly a proxy for pumping and moisture-related loss of support beneath joints, a mechanism the classical equation does not represent explicitly at all. FWD load-transfer efficiency ranks 10th of 14 features: a real, non-trivial contribution (consistent with the ablation result earlier in this chapter) rather than a dominant one, with low LTE (poor joint performance) pushing predicted IRI up, exactly the mechanistic pathway (loss of aggregate interlock or dowel efficiency → faulting → roughness) that Appendix PP’s equation encodes only implicitly through the faulting term alone. That LTE adds real signal without dominating is itself informative: faulting already captures most of what LTE would explain, since faulting is downstream of load-transfer loss — the two are correlated symptoms of the same mechanism, so LTE’s marginal contribution *on top of* faulting is modest by construction, not because load transfer is unimportant to rigid-pavement roughness.

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The two climate variables warrant a more mechanistic reading than “climate matters.” For flexible pavements, mean annual temperature interacts with cracking rather than substituting for it in the SHAP plot (Figure 4.3): its effect is concentrated among records that already carry non-trivial transverse crack length, consistent with the well-established sensitivity of asphalt binder stiffness to temperature — a colder climate embrittles the binder and accelerates thermal (transverse) cracking for a given traffic and structural exposure, so temperature acts as an effect modifier on the crack-length–IRI relationship rather than an independent driver. For rigid pavements, precipitation’s contribution is concentrated among high-faulting records, which is the pattern implied by the classical erosion mechanism underlying pumping: water that infiltrates through joints or shoulders erodes fines from the base or subgrade, and repeated heavy axle loads pump the resulting slurry out from under the slab, progressively reducing support directly beneath the joint and increasing differential deflection across it. This is precisely the moisture-related loss-of-support mechanism that the M-E site factor $SF = \text{Age} \times (1 + FI) \times (1 + P_{200})/10^6$ [2] was constructed to proxy in aggregate rather than represent as a direct interaction with faulting; the SHAP-based model recovers the interaction directly from the joint distribution of precipitation and faulting rather than through a pre-specified multiplicative form.

Notably, the model recovers this structure **without being given the M-E equation’s functional form in advance** — it is discovered from the joint distribution of distress, climate, LTE, and IRI across hundreds of LTPP sections. This agreement with established mechanistic-empirical relationships is a useful check on the SHAP explanation itself: a data-driven attribution that recovers known pavement physics is more credible than one that cannot be reconciled with prior engineering knowledge.

5 Discussion

Six findings merit closer discussion.

Hyperparameter tuning improves performance unevenly across pavement families.

Optuna tuning raised flexible-pavement LightGBM R^2 from 0.333 (an untuned configuration without traffic/LTE imputation) to 0.414, but pooled-rigid R^2 fell slightly, from 0.254 to 0.220, under the same tuning procedure. This is a genuine methodological finding rather than a reporting artifact: with only 179 rigid sections partitioned across a 4-fold group cross-validation objective (roughly 35-45 sections per fold), the tuning objective itself carries substantial variance, and Optuna can select hyperparameters (here, deeper trees, `max_depth = 7`) that fit that noisy objective without generalizing better to the held-out split. Rigid-pavement models built on comparably small LTPP samples would likely benefit from nested cross-validation or a larger outer test set to mitigate this effect.

Falling Weight Deflectometer load-transfer efficiency is a substantively useful addition to the feature set.

Including it improves rigid-pavement accuracy modestly and, more importantly, gives the SHAP analysis a direct mechanistic variable to attribute IRI to, rather than only the faulting symptom of joint-support loss. Of the feature extensions considered in this study, incorporating LTE has the clearest grounding in pavement engineering mechanics.

Ablating traffic loading reframes its role from negligible to modest but genuine.

Excluding AADTT/KESAL costs 0.019 R^2 for flexible pavements and 0.017 R^2 for pooled rigid pavements — well short of the dominant role classical mechanistic-empirical formulations often assign traffic loading, but distinguishable from zero. Because imputation rather than row deletion is used to handle its missingness, this comparison is made on the full analysis sample rather than on a smaller traffic-complete subset that could itself be non-representative.

Forecasting from recent history and structural attribution are distinct tasks with different practical uses, and both are reported.

The operational, lagged-IRI models reach R^2 up to 0.70 for flexible pavements, since a section’s own recent IRI trajectory is a strong predictor of its next reading. The two model families serve different purposes: a pavement management system deciding when to re-inspect a section benefits from the operational model, whereas an agency evaluating which design or maintenance treatment to specify benefits from the structural model, which isolates the effect of distress and structural variables independent of a section’s own measurement history.

Pooling outperforms splitting for rigid pavements at the present sample size, an empirical rather than assumed result (Chapter 4) that directly addresses whether JPCP/JRCP and CRCP should be modeled jointly or separately.

Naive-split inflation scales with how much of the outcome the model fails to explain, not with panel structure alone. Chapter 4 reports a controlled comparison — identical tuned hyperparameters evaluated under a naive and a group-aware split — across all four family/submodel configurations. The inflation ranges from negligible for flexible pavements (0.417 vs. 0.414) to

substantial for pooled rigid pavements (0.440 vs. 0.220) to extreme for the JPCP/JRCP-only submodel (0.355 vs. -0.294), tracking almost exactly the ordering of how little genuine variance each structural feature set explains. This is a more specific and more useful claim than “naive splits inflate accuracy”: it predicts, and the data confirm, that the risk of leakage-driven inflation is greatest precisely where a model’s explanatory power is weakest — the situation in which a naive split’s ability to smuggle in section identity matters most. A weaker structural model is therefore not only less accurate on its own terms; it is also more vulnerable to being mistaken for a stronger one under an insufficiently careful evaluation protocol.

Beyond these six points, the gap between single-split and cross-validated R^2 (0.414 vs. 0.227 for flexible pavements) illustrates that panel data with repeated section visits is particularly unforgiving of evaluation protocols that do not account for its structure. This has implications beyond the present study: work using LTPP or comparable longitudinal pavement datasets should report train/test splitting strategy in sufficient detail for this source of optimism bias to be assessed by readers. Two further robustness checks (Chapter 4) support the structural model’s headline numbers: a five-draw multiple-imputation ensemble reproduces the single-imputation results to within 0.05 R^2 for every family, and results are essentially unchanged across three modern gradient-boosting implementations, indicating that neither the imputation strategy nor the specific choice of ensemble library is responsible for the reported accuracy levels.

6 Limitations

This study has the following limitations, which should be considered when interpreting its findings:

- **Complete-case selection on structural predictors.** Both panels retain only records with a fully observed *structural* core; traffic and LTE are imputed rather than excluded, but the structural core itself (distress, climate, thickness) is still restricted to complete cases. Sections with sparser instrumentation are systematically excluded, and whether they differ systematically from the retained sample has not been formally tested.
- **Mixed age provenance.** 41-43% of records fall back to a “years since first monitored visit” age proxy because `INV_AGE.CONSTRUCTION_DATE` is unavailable for that section; this is flagged in the data (`AGE_SOURCE` column). We attempted to test its effect directly by refitting the structural model on the verified-construction-date subset alone (Chapter 4), but that subset retains only 19-61 sections per family — too few for a stable estimate, and the resulting R^2 values are not interpretable as evidence either way. Resolving this properly would require either a larger verified-age subsample or an explicit model of age-source-related measurement error, neither of which is attempted here.
- **Imputation uncertainty is largely, but not completely, addressed.** A five-draw stochastic multiple-imputation check (Chapter 4) shows the single-imputation `IterativeImputer` pipeline used throughout reproduces held-out R^2 within 0.003-0.05 across all four families, which is reassuring but does not amount to a formal Rubin’s-rules variance decomposition of a fitted coefficient; no such decomposition is attempted for a black-box ensemble model in this study.
- **Rigid submodel instability, and pooling as a sample-size-driven choice rather than a permanent one.** The JPCP/JRCP-only and CRCP-only submodels are informative for the pooling-versus-splitting comparison but are themselves too small (61-118 sections) to be deployed as standalone operational models; their cross-validation variance is correspondingly large. The pooled rigid-pavement model is therefore adopted because it is the better-performing choice *at LTPP’s current rigid-pavement sample size*, not because JPCP/JRCP and CRCP are assumed to share identical deterioration mechanics. As additional LTPP monitoring cycles accumulate and sub-family sample sizes grow, the pooling-versus-splitting comparison in Chapter 4 should be revisited, and separate models may become preferable.
- **Sample size**, while comparable in order of magnitude to [3]’s flexible-only 2,699-record study, is smaller than that benchmark for the flexible family and reflects only LTPP’s current curated analysis-ready tables rather than the full historical survey record.
- **This study uses a single snapshot of LTPP data (Standard Data Release 39, September 2025).** LTPP data is periodically revised; a longitudinal reproducibility check against subsequent releases is left for future work.

7 Conclusion

Rigid pavements have remained on the periphery of machine-learning research on IRI prediction, not because they are of lesser practical importance, but because no prior study has extended the explainable ensemble-learning approach developed for asphalt pavements to JPCP, JRCP, and CRCP under a comparable, jointly benchmarked framework. This study closes that gap using a reproducible pipeline built directly against the FHWA LTPP Standard Data Release, spanning 1,552 flexible-pavement and 1,693 rigid-pavement records across the U.S. and Canadian LTPP network.

An Optuna-tuned gradient-boosted ensemble, evaluated under a section-grouped train/test split that prevents repeated visits to the same physical pavement from inflating reported accuracy, explains 41.4% of out-of-sample IRI variance for flexible pavements and 22.0% for pooled rigid pavements using distress, structure, and climate variables alone, rising to 69.8% and 39.1%, respectively, when each section’s own recent IRI history is included as an additional predictor. In both pavement families, the ensemble model substantially outperforms a mechanistic-empirical-style linear regression and trains several times faster than a comparably accurate random forest. SHAP-based interpretation recovers, without being given the functional form of any mechanistic-empirical equation, the same distress mechanisms — transverse cracking and patching for flexible pavements, joint faulting and load-transfer efficiency for rigid pavements — that decades of pavement engineering research established by hand, while additionally identifying climate interactions that fixed-coefficient equations cannot represent.

Four further results have direct implications for how LTPP data is used in pavement performance modeling. First, evaluation protocol affects reported accuracy as much as model choice, and its effect is not uniform: a controlled comparison holding hyperparameters fixed shows that a train/test split ignoring LTPP’s repeated-visit panel structure inflates R^2 by an amount that scales with how much genuine variance the structural features fail to explain — negligibly for flexible pavements, substantially for pooled rigid pavements, and by more than half a point of R^2 for the least-explained rigid submodel. Reporting splitting strategy in sufficient detail for this to be assessed should be standard practice in LTPP-based machine-learning studies. Second, traffic loading and load-transfer efficiency each contribute measurably but modestly to model accuracy once properly imputed, indicating that neither should be excluded for missingness nor treated as the dominant explanatory variable. Third, at the sample sizes currently available in LTPP, a single pooled rigid-pavement model spanning JPCP, JRCP, and CRCP — distinguished by a pavement-type indicator — generalizes better than separate models fit to each sub-family, despite their differing distress mechanisms, though this is treated as a sample-size-driven choice to be revisited as more rigid-pavement data accumulates rather than a claim that the sub-families share identical deterioration mechanics. Fourth, the reported accuracy is not an artifact of a particular gradient-boosting library or imputation procedure: results are essentially unchanged across three modern gradient-boosting implementations and across single- versus multiple-imputation treatment of missing traffic and load-transfer data.

These findings have practical relevance for both highway agencies and the pavement engineering research community. Agencies prioritizing maintenance across mixed flexible and rigid networks can apply a common modeling framework rather than maintaining separate tools built on incompatible assumptions, selecting between the structural and operational model variants according to whether the underlying question is diagnostic (identifying what is driving roughness) or predictive (determining when a section should next be inspected). For the research community, the complete pipeline — from raw data acquisition through model training and SHAP explanation — is released as open-source code, allowing the reported results to be verified, extended to future LTPP releases, and adapted to additional structural or sensor-derived features. The limitations discussed in Chapter 6, particularly the sample size available for rigid pavements and reliance on a single LTPP release, motivate a natural direction for future work: revisiting this framework as additional LTPP monitoring cycles and Falling Weight Deflectometer coverage accumulate, and incorporating structural response data beyond load-transfer efficiency, such as full deflection basins, as a further mechanistic input to explainable IRI prediction.

7.1 Data and Code Availability

The complete reproducible pipeline (data acquisition, cleaning, modeling, and this manuscript) is available on GitHub at https://github.com/sabernaserlavi-60/2026_International-Roughness-Index-Prediction-Using-the-LTPP under an open license. Raw LTPP data is public domain, distributed by FHWA InfoPave.

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